

MMU753
TH#3
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PART-I

$$(2) m_{eq} \frac{dv}{dt} = F_x - mg \sin \theta - f + g \cos \theta - 0.5 \rho A C_d (v + v_w)^2$$

Equivalent mass due to rotating parts;

$$E_{kin, trans} = \frac{1}{2} m v^2 \quad E_{kin, rot} = \frac{1}{2} \sum I_{cm} \omega_w^2 + \frac{1}{2} I_e \left(\frac{G, R}{r} v \right)^2$$

$$E_L = \frac{1}{2} \left[m + \frac{\sum I_{cm}}{r^2} + \frac{I_e (G, R)^2}{r^2} \right] v^2$$

$\underbrace{\hspace{10em}}_{m_{eq}}$

If we call additional term as 'd', $m_{eq} = m(1+d)$

(3) Linearizing NDE in (1) around point (θ_0, F_0, u_0) ;

Get Jacobians:

$$\frac{\partial f}{\partial u} = \frac{-\rho A C_d (u_0 + v_w)^2}{m_{eq} u}$$

$$\frac{\partial f}{\partial F} = \frac{1}{m_{eq} v} \quad \frac{\partial f}{\partial \theta} = \frac{m}{m_{eq} v} \cdot (-g \cos \theta_0 + f \sin \theta_0)$$

$$\dot{u} \cdot u' = \frac{\partial f}{\partial u} \cdot u' + \frac{\partial f}{\partial F} \cdot F' + \frac{\partial f}{\partial \theta} \cdot \theta'$$

$$\dot{u} \cdot u' = \frac{-\rho A C_d (u_0 + v_w)^2}{m_{eq} v} \cdot u' + \frac{1}{m_{eq} v} \cdot F' + \frac{m}{m_{eq} v} (-g \cos \theta_0 + f \sin \theta_0) \theta'$$

$$m_{eq} \cdot \frac{\dot{u} \cdot u'}{\rho A C_d (u_0 + v_w)} + u' = \frac{1}{\rho A C_d (u_0 + v_w)} \cdot F' + \frac{m}{m_{eq} v} \cdot \frac{(-g \cos \theta_0 + f \sin \theta_0)}{\rho A C_d (u_0 + v_w)} \cdot \theta'$$

Rearrange into standard form:

$$\boxed{\tau \ddot{u}' + u' = K(F' + d)}; \quad \boxed{\tau = \frac{m_{eq} v}{\rho A C_d (u_0 + v_w)}} \quad \boxed{K = \frac{1}{\rho A C_d (u_0 + v_w)}}$$

Input

(1)

$$\boxed{d = mg(-\cos \theta_0 + f \sin \theta_0)} \quad \text{Disturbance}$$

③ (continued)

As depicted, F' is input and d is distance.

Following parameters are chosen for system:

$$u_0 = 40 \text{ m/s} \quad \theta_0 = 0 \quad m = 1250 \text{ kg} \quad \rho = 1.225 \text{ kg/m}^3$$

$$A = 1.2 \text{ m}^2 \quad Cd = 0.5 \quad f = 0.015 \quad u_w = 1.5 \text{ m/s}$$

$$\sum I_w = 4 \text{ kg.m}^2 \quad I_e = 0.15 \text{ kg.m}^2 \quad r = 0.3 \text{ m}$$

$$D = 3.0 \quad G = 1.1 \text{ (4th gear)}$$

$$m_{\text{eqv}} = m + \frac{\sum I_w}{r^2} + \frac{I_e (G \cdot r)^2}{r^2}$$

$$m_{\text{eqv}} = (1250) + \frac{4}{(0.3)^2} + \frac{(0.15) (1.1 \cdot 0.3)^2}{(0.3)^2}$$

$$m_{\text{eqv}} = 1313.1 \text{ kg}$$

$$\bar{t} = \frac{m_{\text{eqv}}}{\rho A C_d (u_0 + u_w)} = \frac{(1313)}{(1.225)(1.2)(0.5)(40 + 1.5)}$$

$$\boxed{\bar{t} = 42.53 \text{ s}}$$

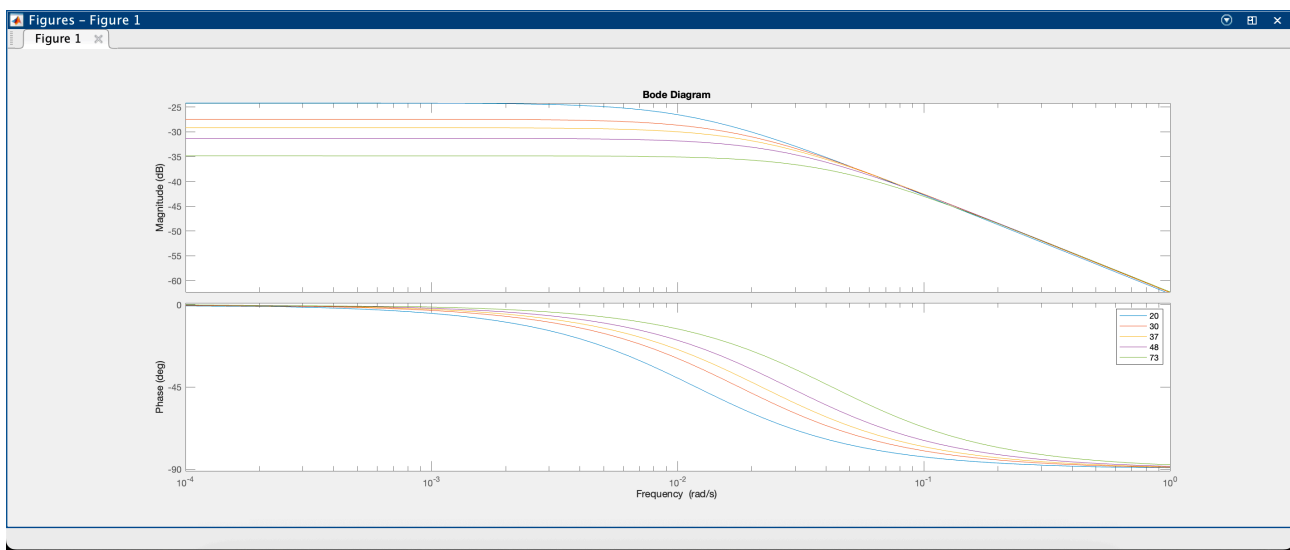
PART-I

(5) For ICE, the input given to the system, (F_x), is throttle position.

For Electric motor, it is the electrical current.

PART-I

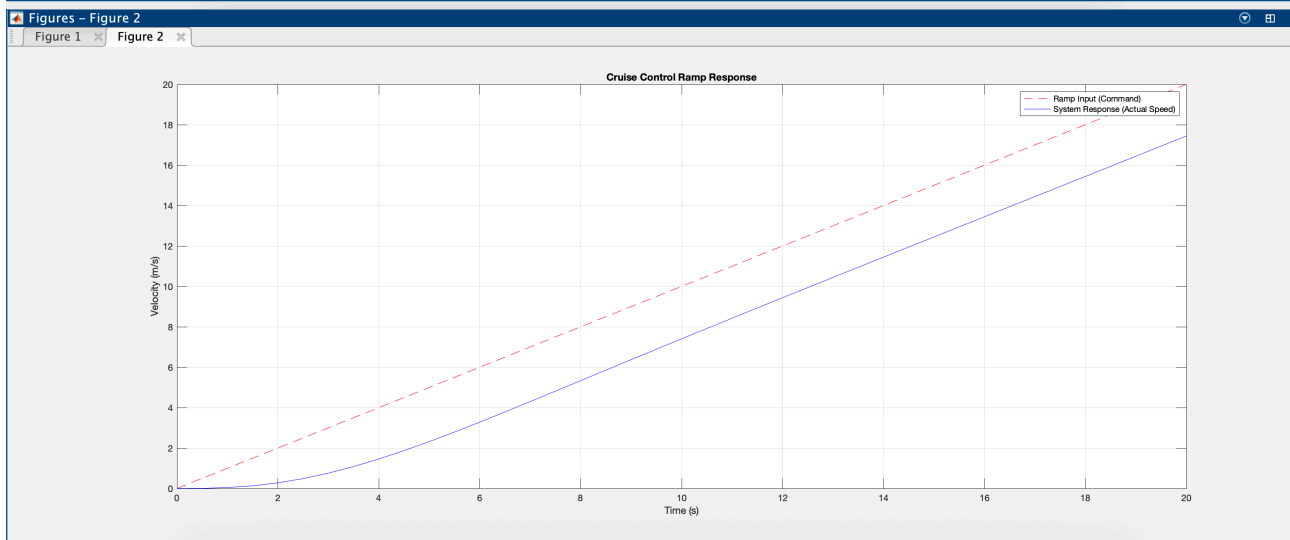
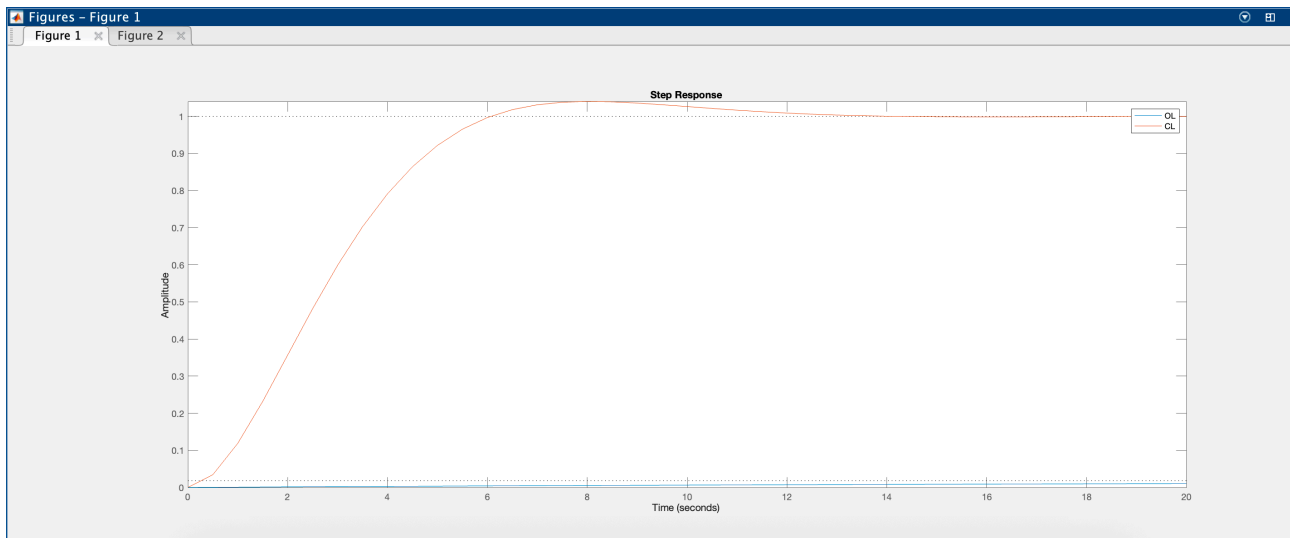
4) It can be observed that as velocity increases DC Gain (value of gain plot on low frequencies) get's lower, which means as drag loss is proportional to velocity, in high speeds the car is more insensitive to force inputs coming from engine as it needs more force because of the drag force. Another observation is that as velocity increases, cornering frequency shifts to the right, to higher frequency values. This is due to fact that as velocity increases time constant decreases.



PART-I

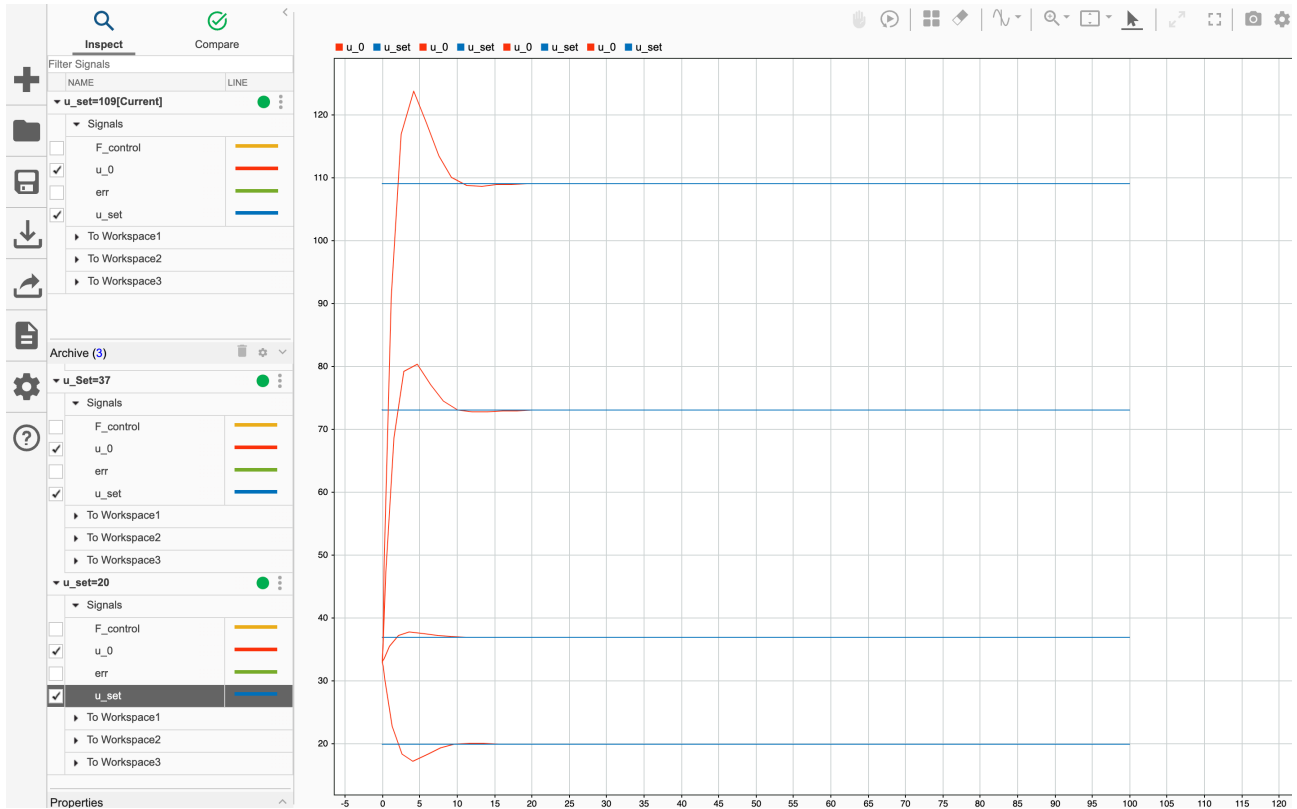
6) Following are the step and ramp responses of the closed-loop system with PI Controller constructed using pole placement technique, with maximum percent overshoot is smaller than %4 and settling time is smaller than 10 seconds.

It can be seen that the system can track the step response with zero steady-state error. Nevertheless it can track the ramp input with a steady-state error. This can be explained due to the concept of system types, where the system in the question is Type-1, which can track a step input with zero steady-state error and a ramp input with a non-zero steady-state error. As the system has only one free integrator due to Integral control action, the system is Type-1.



PART-I

7) It can be seen that as difference between the linearization point and test point increase, the overshoot and settling time increases. This shows that even though the model shows similar performance in nonlinear system at the linearization point, at points away from it the performance decreases, as expected.



PART-I

8)

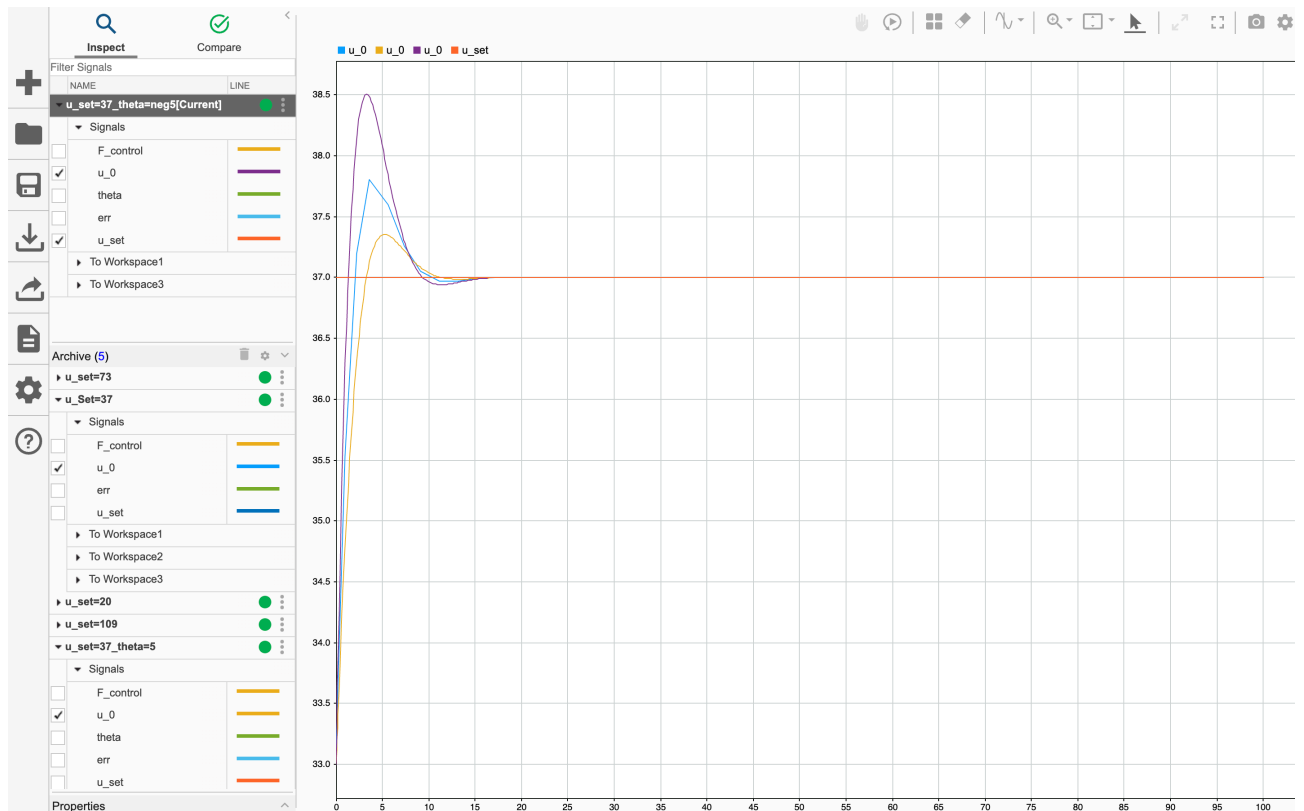
Disturbance (road slope) is defined as step input starting at time zero. It is tried for two values: $\theta=7$ degrees and $\theta=-7$ degrees. In the below plot

Purple: 7 degrees road slope

Blue: No slope

Yellow: -7 degrees road slope

It can be seen that overshoot is decreased for the positive slope case. This is because now the system fights against gravity also, which adds additional damping. Contrary, for the case where the slope is negative the damping decreases and overshoot increases.

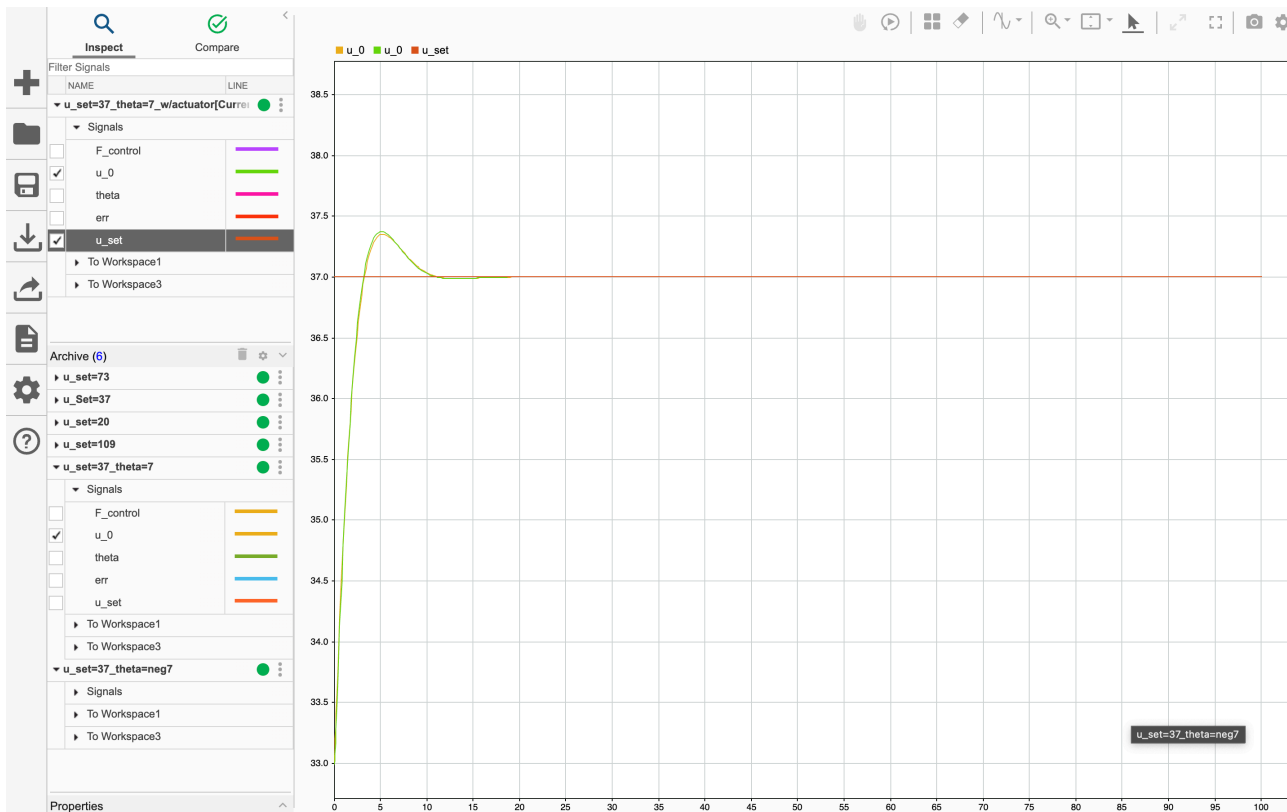


PART-I

9)

For actuator, gain is selected as 1 and time constant is selected as 0.05 seconds. As time constant of the actuator is very small compared to the system's time constant (which is bigger than 20 seconds in the scenario where it is smallest), output is not effected much and can be neglected.

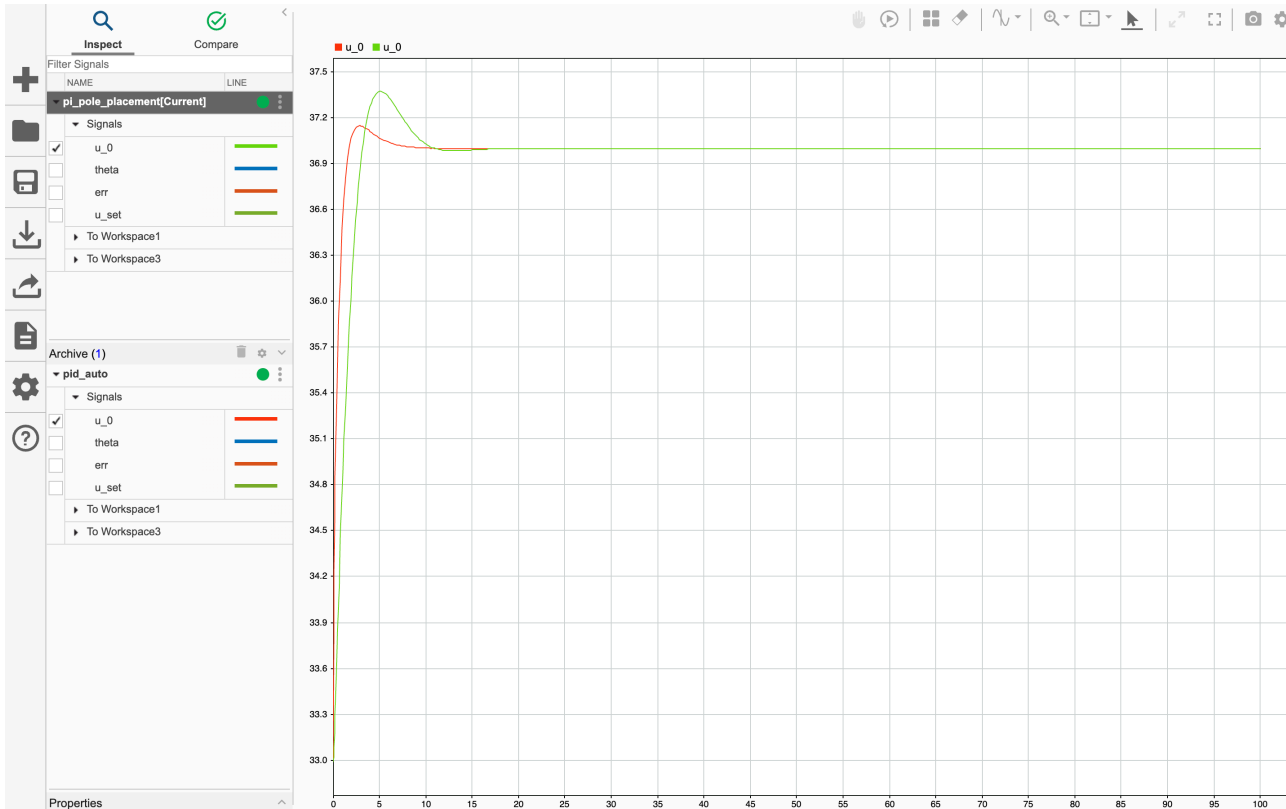
Note that scenario is run at 37 m/s set speed and with disturbance is positive 7 degrees (constant).



PART-I

10)

Using PID block and PID tuner app, PID parameters are set. In graph, red is response with PID controller and green is the PI controller constructed using pole placement. It can be observed that both overshoot and settling time are improved when PID and auto tuner are used.



11) Using PID controller is more advantageous compared to PI controller in following senses:

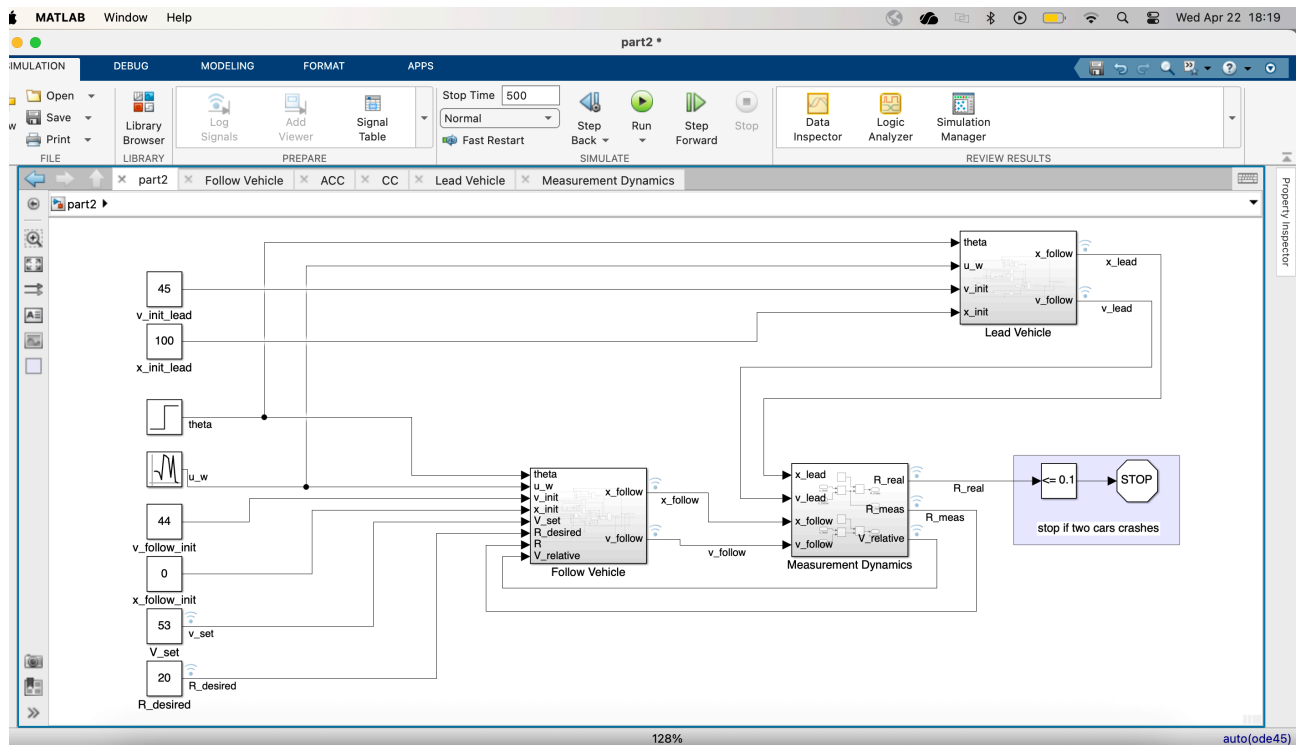
- D term allows using bigger P terms for faster response, while keeping overshoot small thanks to the braking behavior of D term.

Nevertheless, in real-world applications the inherent noise leads instability of D term, because of that generally PI controller is used.

PART-I

12)

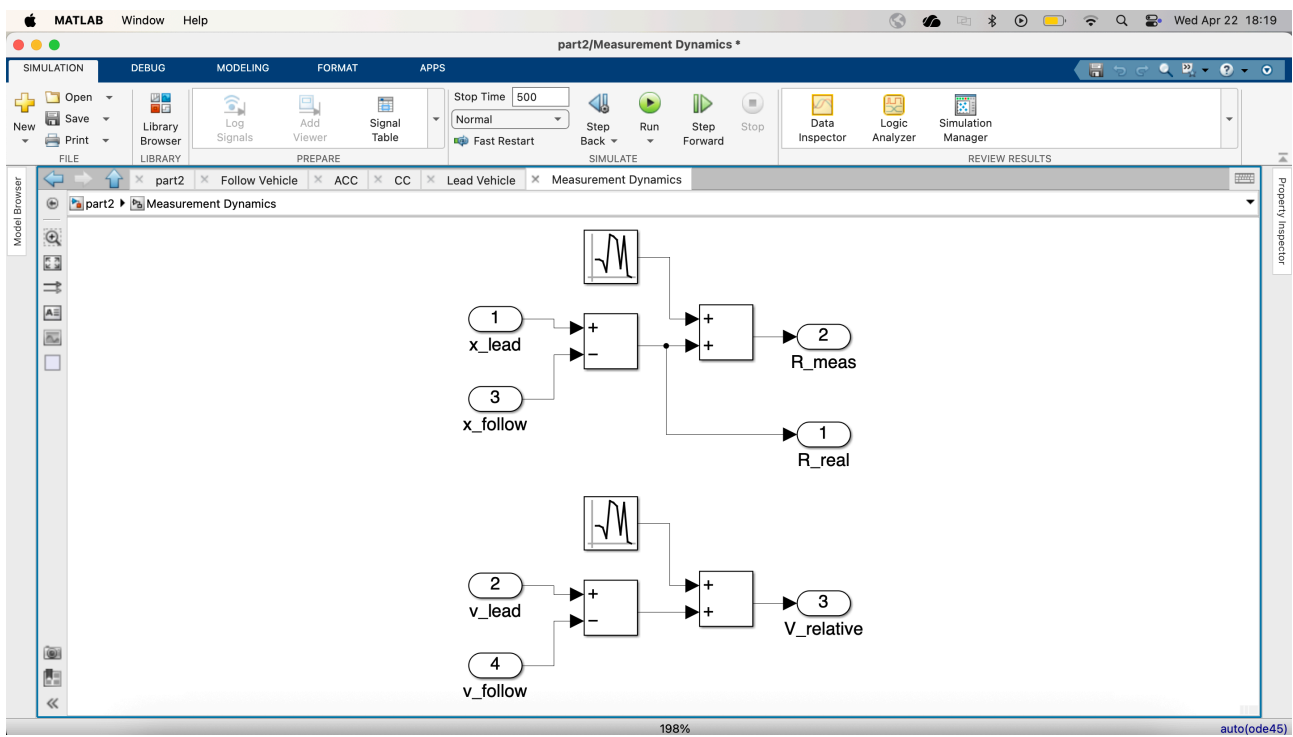
Following Simulink model is constructed for simulations.



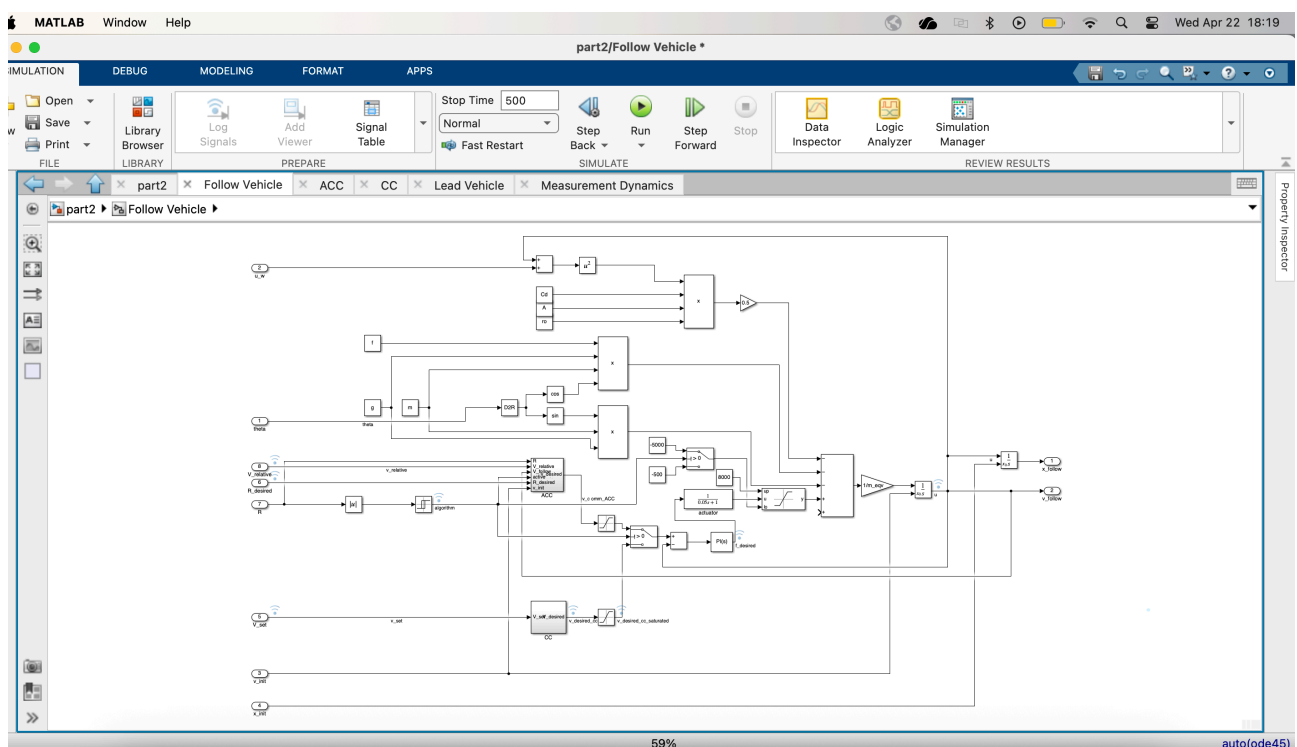
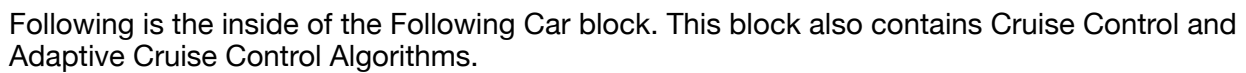
It can be seen that two cars are modeled as follow and lead vehicles. Initial velocities and positions are defined for both of them. Also as they are at the same road, same wind and road slope data are fed to both of them. Note that for a more realistic simulation, following car should be receiving this inputs by a time delay.

In reality, measurements are not perfect. So, Range measurement of follow car and relative car velocities are perturbed using normally distributed variables. For Range, used variable has 0 mean and 0.2 variance and updated at 10 Hz, and for relative velocity mean is zero and variance is 0.1, also updated at 10 Hz. Following is the inside of the Measurement Dynamics block.

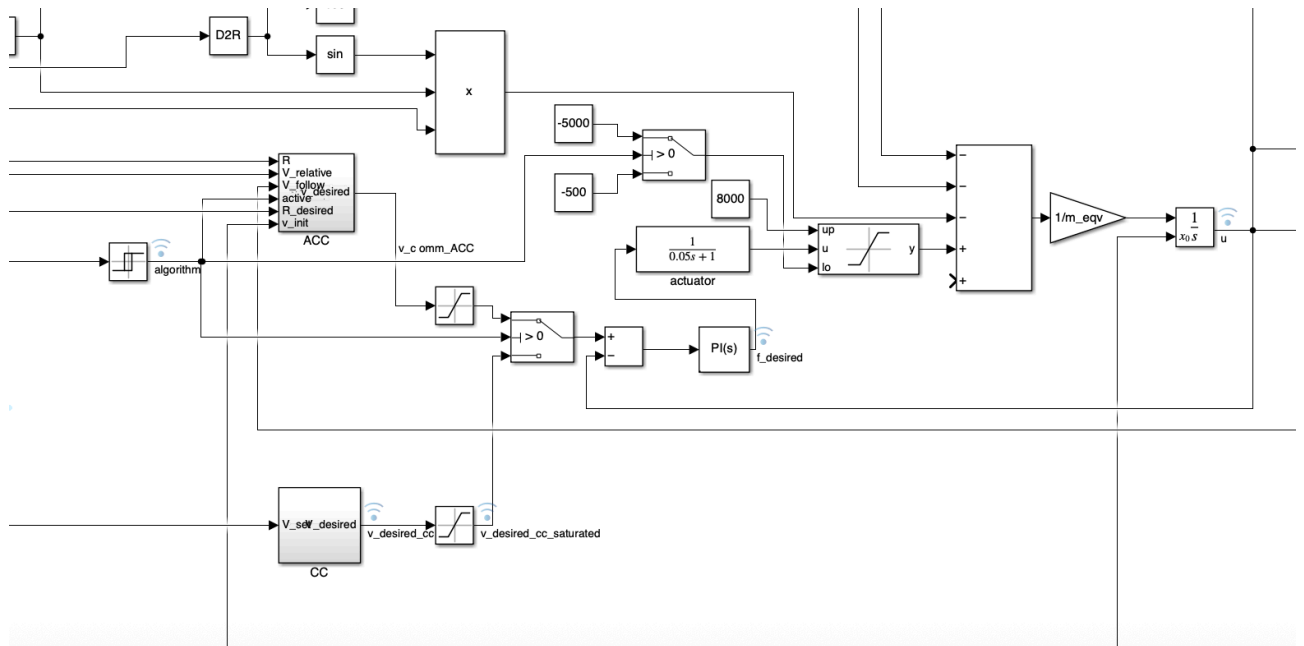
For wind, a disturbance where mean is zero and variance is 2 m/s is used. For road gradient, 3 degrees positive slope is used.



Following is the inside of lead vehicle. It is the same model used in Part-1 Question 4. Same parameters are used for both cars, nevertheless they can easily be changed.



For a more detailed look, control algorithm is zoomed:

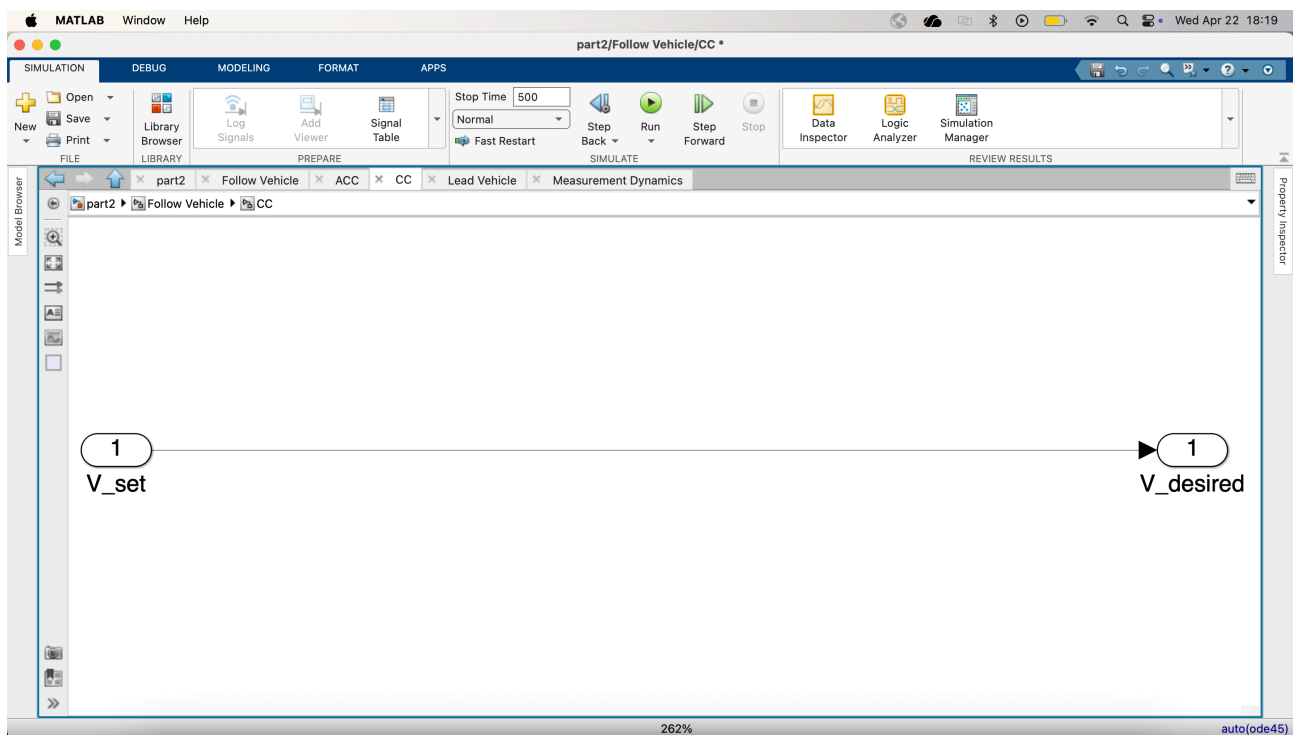


Starting from right; it can be seen that all forces are summed and divided to equivalent mass, then integrated and velocity is obtained. There is a control force input to the system which is filtered through a dynamic saturation block, where for CC only engine brake is allowed and for ACC normal braking is also allowed. For this reason as maximum limit of saturation block is same for both (8.000), minimum limit is -500 for CC and -5000 for ACC.

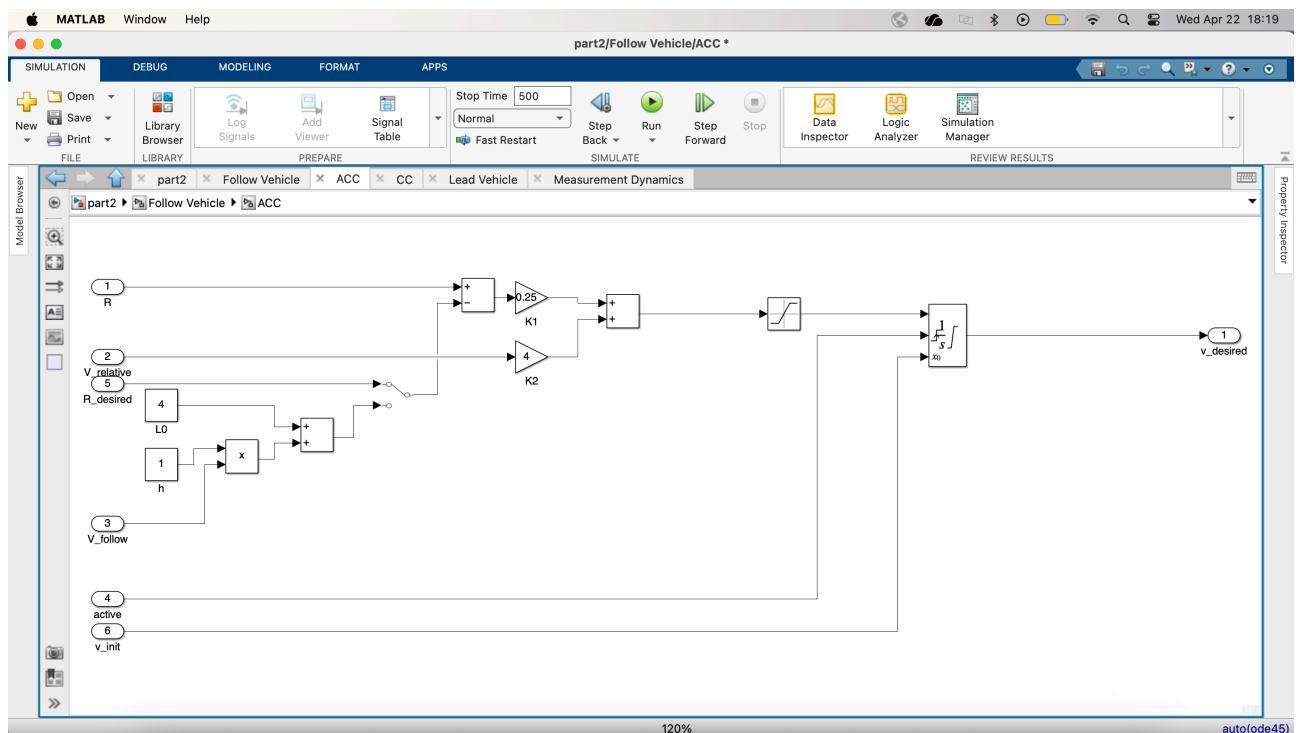
It can be seen that there is a signal named “algorithm”. This signal decides which algorithm is active. For a more robust algorithm, a trick is applied to this signal, where a deadband is created between 45m and 30m, which mean in order for car to switch to CC, range should drop below 20m (desired range) and in order to switch to ACC, range should rise above the 35m. This makes the algorithm more robust as it allows some oscillations and overshoots.

It can be seen that actuator is modeled as a first order file with 0.05 seconds time constant. Both the CC and ACC algorithms outputs a desired velocity. This velocity is saturated again between 0 m/s and 50 m/s and subtracted with actual velocity and velocity error is obtained. This error is fed into a PI controller and the desired force is created.

Following is the inside of the CC block, in which the set velocity directly fed as desired velocity.

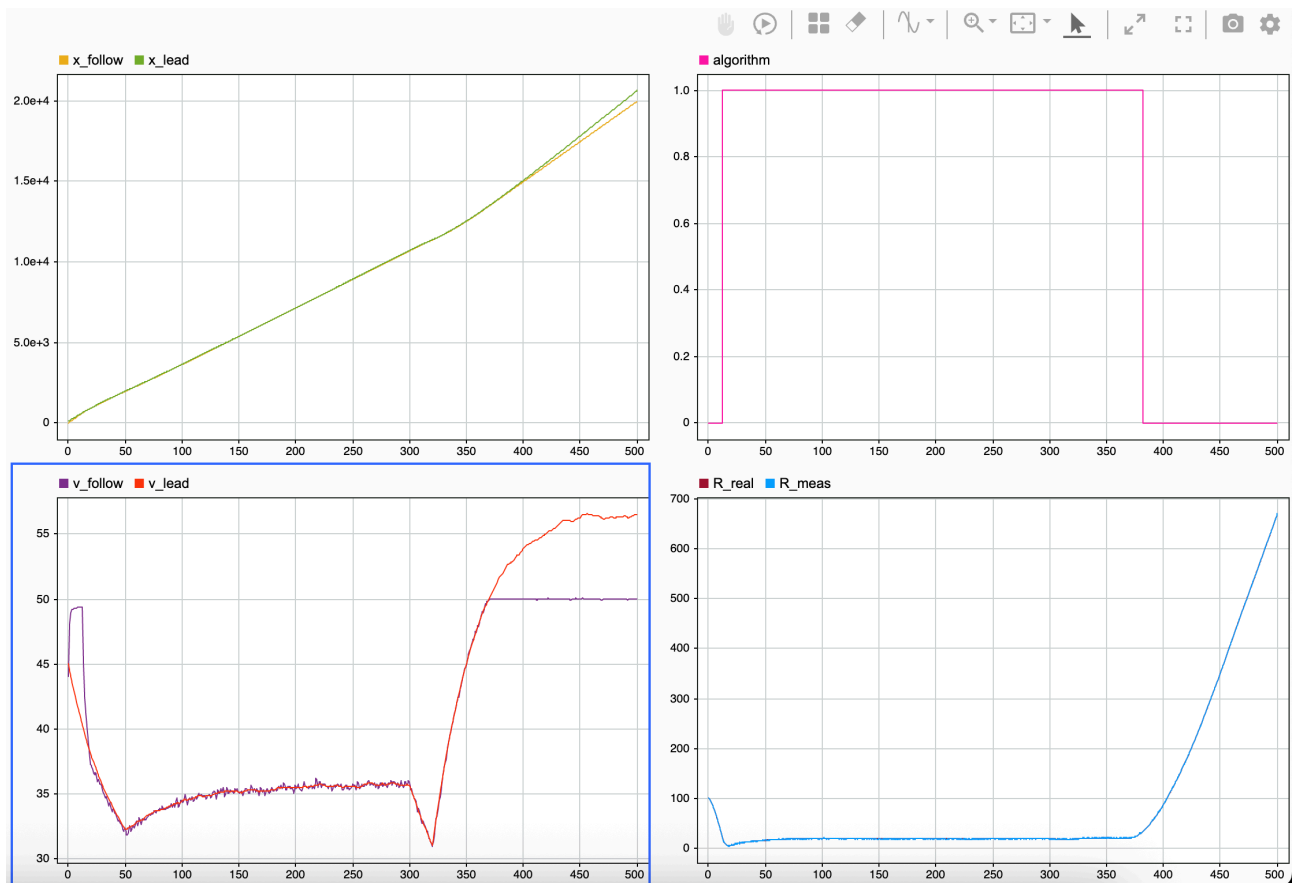


Following is the inside of the ACC block. It can be observed that both Type-1 and Type-2 controllers are implemented and can be switched with manual switch block. Output of these algorithms is the desired acceleration, in which filtered through a saturation block with upper limit 2.5 m/s^2 and lower limit -1 m/s^2 . After that this acceleration is integrated and desired velocity is obtained.

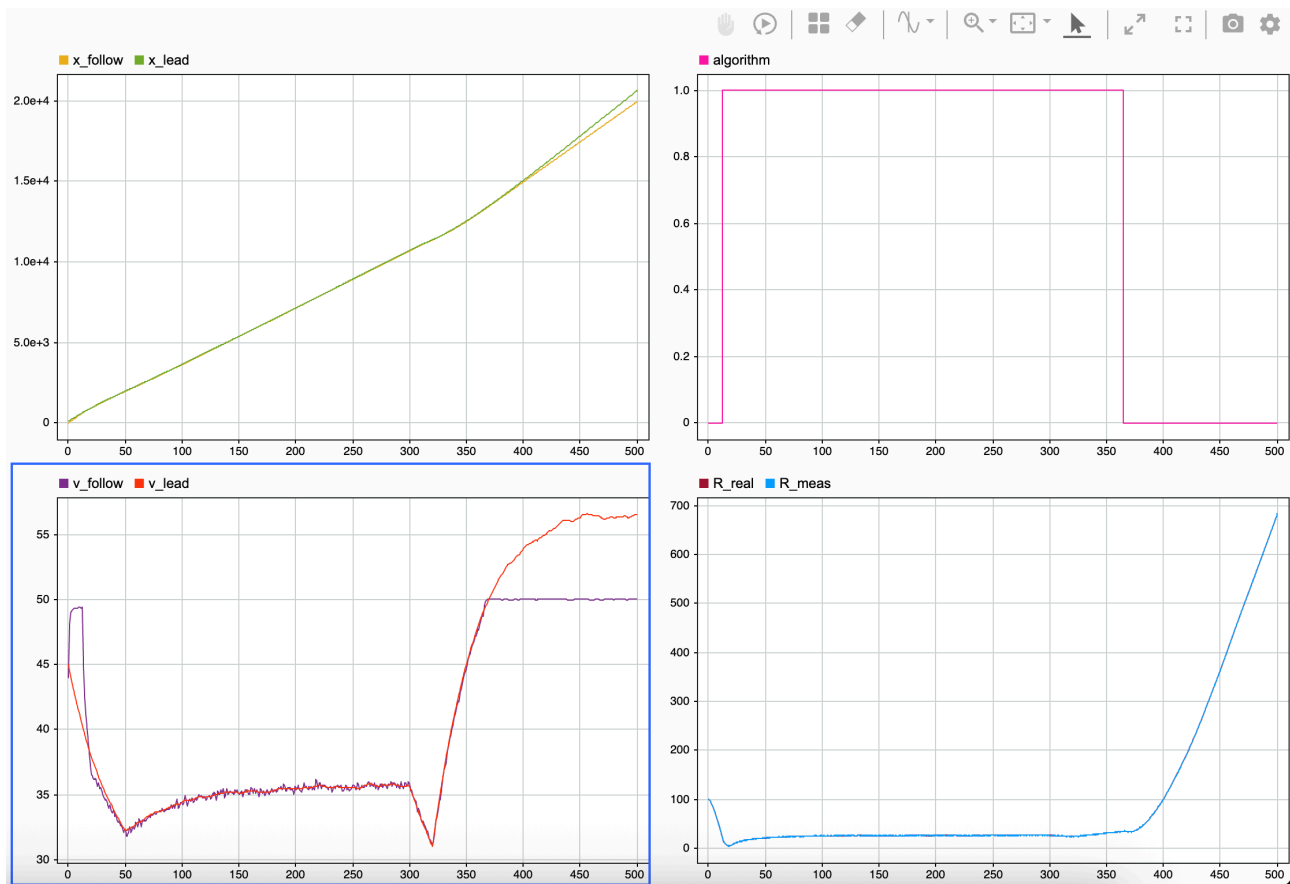


First, a simulation is ran with Type-1 controller. Following are the results. There are four different plots here. At top left, forward distances the cars are covering can be observed. At top right, the mode in which the algorithm works is visualized, where 0 means CC and 1 means ACC. At bottom left follow and lead car velocities are given. At bottom right, measured and real range (distance between cars) values are visualized.

It can be seen that at the beginning there is 100m distance between two cars, so following var works on CC mode. Nevertheless, as following car's initial velocity is larger and lead car starts braking, the distance between them closes and at time around 12, ACC mode becomes active. Up until 380 seconds, following car keeps the distance at 20 meters with some oscillations, mostly caused by measurement noise. Also it can be observed that in order to that following var aligns it's velocity to lead car's. At time 380 seconds, Lead Car starts accelerating and increases the distance between them, when the distance becomes larger than 35m, Following Car switches into CC mode and reaches it's set velocity, 50 m/s (note that in block diagram it's set as 53 m/s by mistake, but there is already a limit of 50 m/s for it, so steady-state velocity is 50 m/s).



14) Similar things can be said for Type-2 controller as the result is very similar.



16) For reference example from Mathworks, a sinusoidal velocity profile is used for Lead Car. It can be seen that the motion of Ego car is smooth. As structure of MPC allows implicit integration of constants such as acceleration and braking forces, engines powers etc., the problem setup is much cleaner and design process is easier.

